

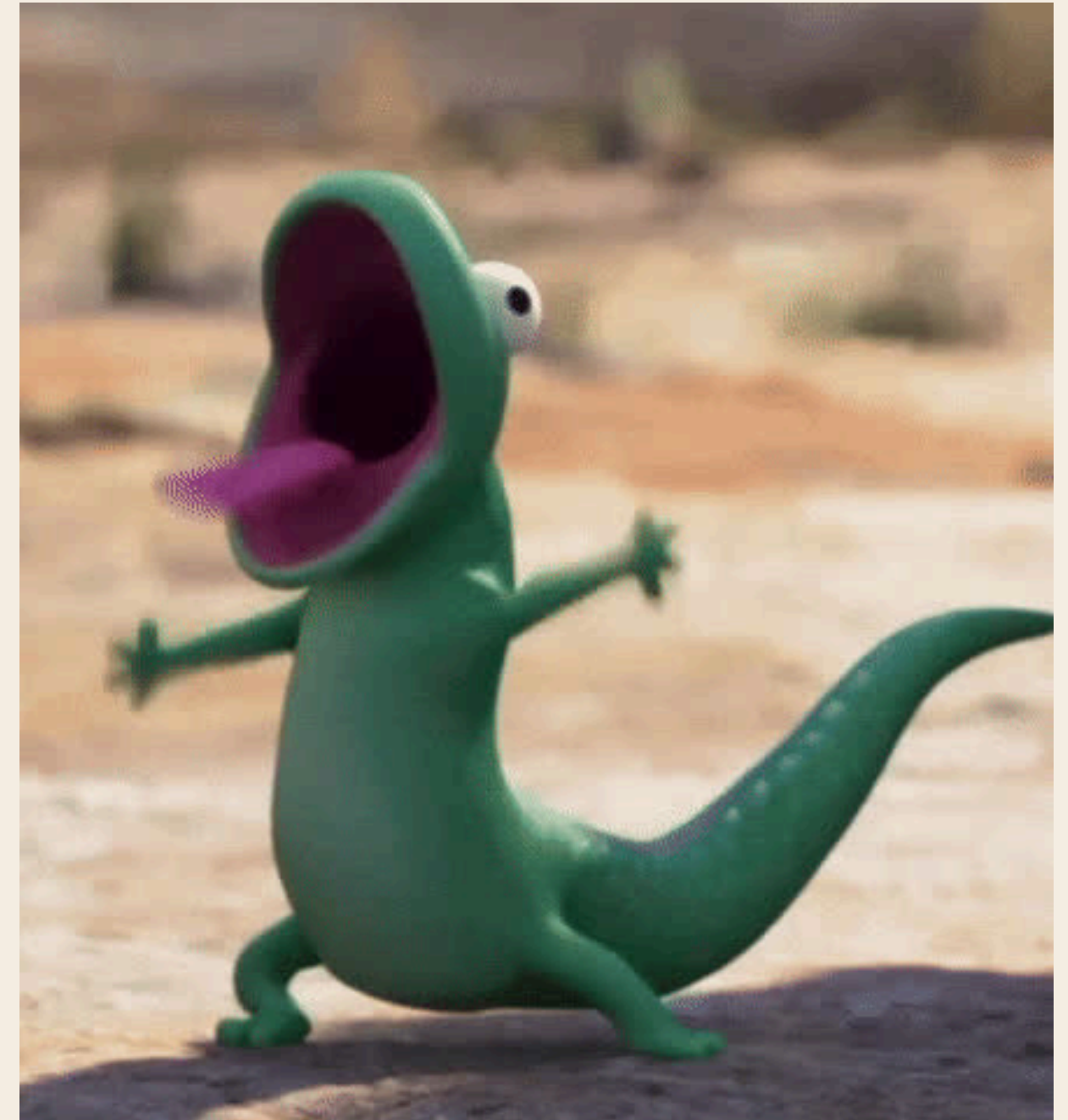
DSA1101

Introduction to Data Science

Week 9 Tutorial 7

TODAY'S AGENDA

1. Quick Revision
2. On-site Questions Attempt
3. On-site Questions Discussion
4. Off-site Questions Discussion



QUICK REVISION

NAIVE BAYES

Supervised machine learning algorithm used for classification, based on
Bayes' theorem

‘BAYES’

Bayes Theorem:

$$\mathbf{P(A|B)} = \frac{\mathbf{P(B|A) P(A)}}{\mathbf{P(B)}}$$

‘NAIVE’

Independence between features

$$\begin{aligned} &P(X_1 = x_1, X_2 = x_2, \dots, X_m = x_m | Y = y_j) \\ &= P(X_1 = x_1 | Y = y_j) P(X_2 = x_2 | Y = y_j) \dots P(X_m = x_m | Y = y_j) \\ &= \prod_{i=1}^m P(X_i = x_i | Y = y_j). \end{aligned}$$

$$P(A \cap B) = P(A)P(B)$$

NAIVE BAYES

In your lecture, you saw the process of:

$$\begin{aligned} P(Y = y_j | X) \\ = \frac{P(X_1 = x_1, X_2 = x_2, \dots, X_m = x_m | Y = y_j) \times P(Y = y_j)}{P(X_1 = x_1, X_2 = x_2, \dots, X_m = x_m)} \end{aligned}$$

$$\begin{aligned} P(X_1 = x_1, X_2 = x_2, \dots, X_m = x_m | Y = y_j) \\ = P(X_1 = x_1 | Y = y_j) P(X_2 = x_2 | Y = y_j) \dots P(X_m = x_m | Y = y_j) \\ = \prod_{i=1}^m P(X_i = x_i | Y = y_j). \end{aligned}$$

+

$$P(X_1 = x_1, X_2 = x_2, \dots, X_m = x_m)$$

=

$$P(Y = y_j | X) \propto P(Y = y_j) \times \prod_{i=1}^m P(X_i = x_i | Y = y_j)$$

NAIVE BAYES FOR CLASSIFICATION

$$P(Y = y_j | X) \propto P(Y = y_j) \times \prod_{i=1}^m P(X_i = x_i | Y = y_j)$$

POV: I want to predict class Y given X features

Final Prediction: Compute score for every class and pick the largest.

$$P(Y = y_1 | X) \propto P(Y = y_1) \cdot \prod P(X_i | Y = y_1)$$

$$P(Y = y_2 | X) \propto P(Y = y_2) \cdot \prod P(X_i | Y = y_2)$$

$$P(Y = y_3 | X) \propto P(Y = y_3) \cdot \prod P(X_i | Y = y_3)$$

NAIVE BAYES IN R

Using library e1071

```
# install.packages("e1071")  
library(e1071)
```

Calling model:

```
model = naiveBayes(y ~ x, data = data)
```

Making predictions to new data:

```
preds <- predict(model, test_data)
```

EVALUATING NAIVE BAYES

our fav confusion matrix!

	Actually Positive (1)	Actually Negative (0)
Predicted Positive (1)	True Positives (TPs)	False Positives (FPs)
Predicted Negative (0)	False Negatives (FNs)	True Negatives (TNs)

Sometimes Negative and Positive will switch places. So please read carefully before answering!

Accuracy =

$$(TP + TN) / (TP + TN + FP + FN)$$

$$TPR (Recall) = TP / (TP + FN)$$

$$FPR = FP / (FP + TN)$$

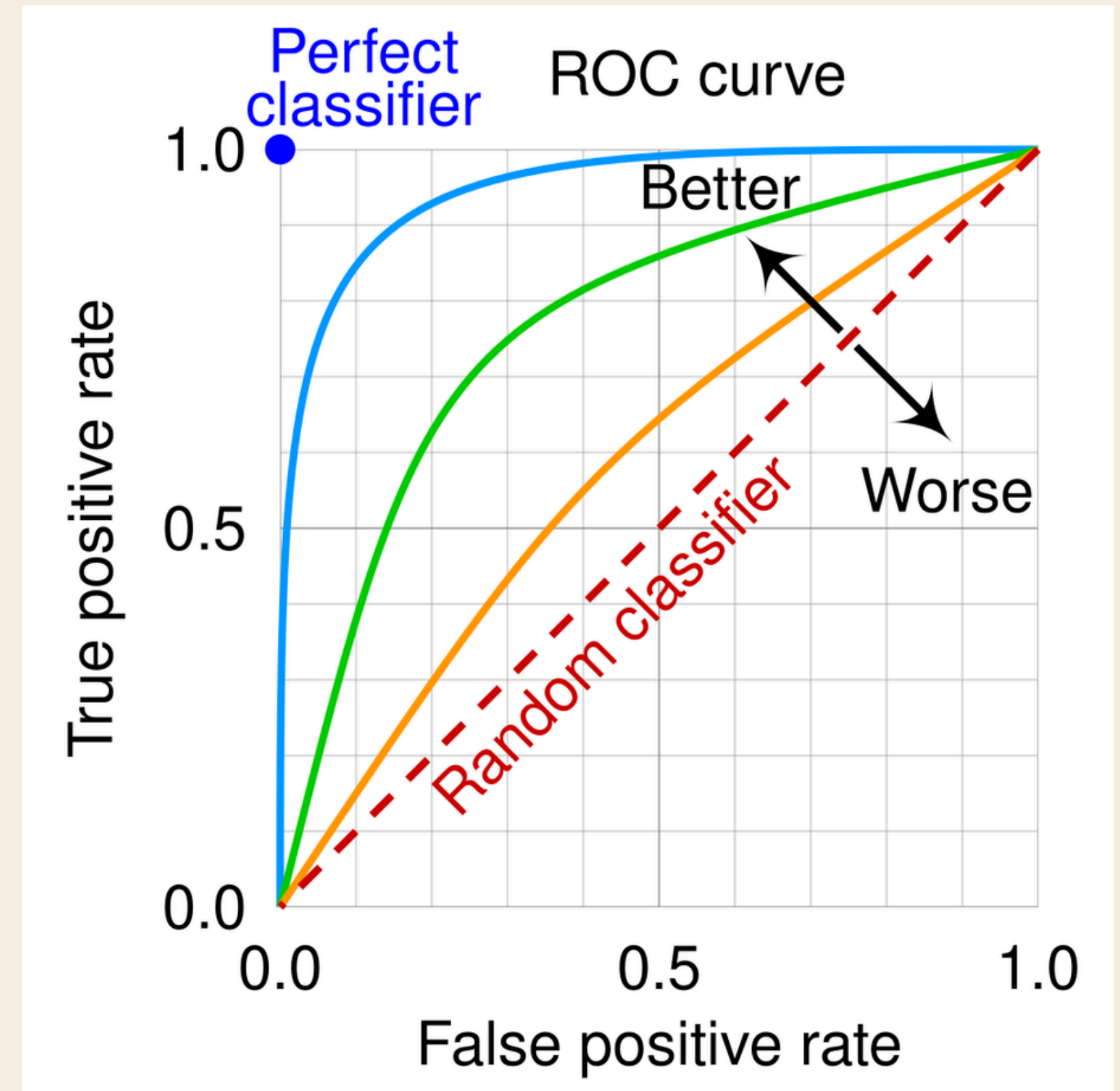
$$FNR = FN / (TP + FN)$$

$$Precision = TP / (TP + FP)$$

NEW EVALUATION TECHNIQUE

ROC Curve

ROC curve plots the TPR and FPR from your confusion matrix at every possible threshold, showing how the tradeoff between them changes as you move the classification cutoff.





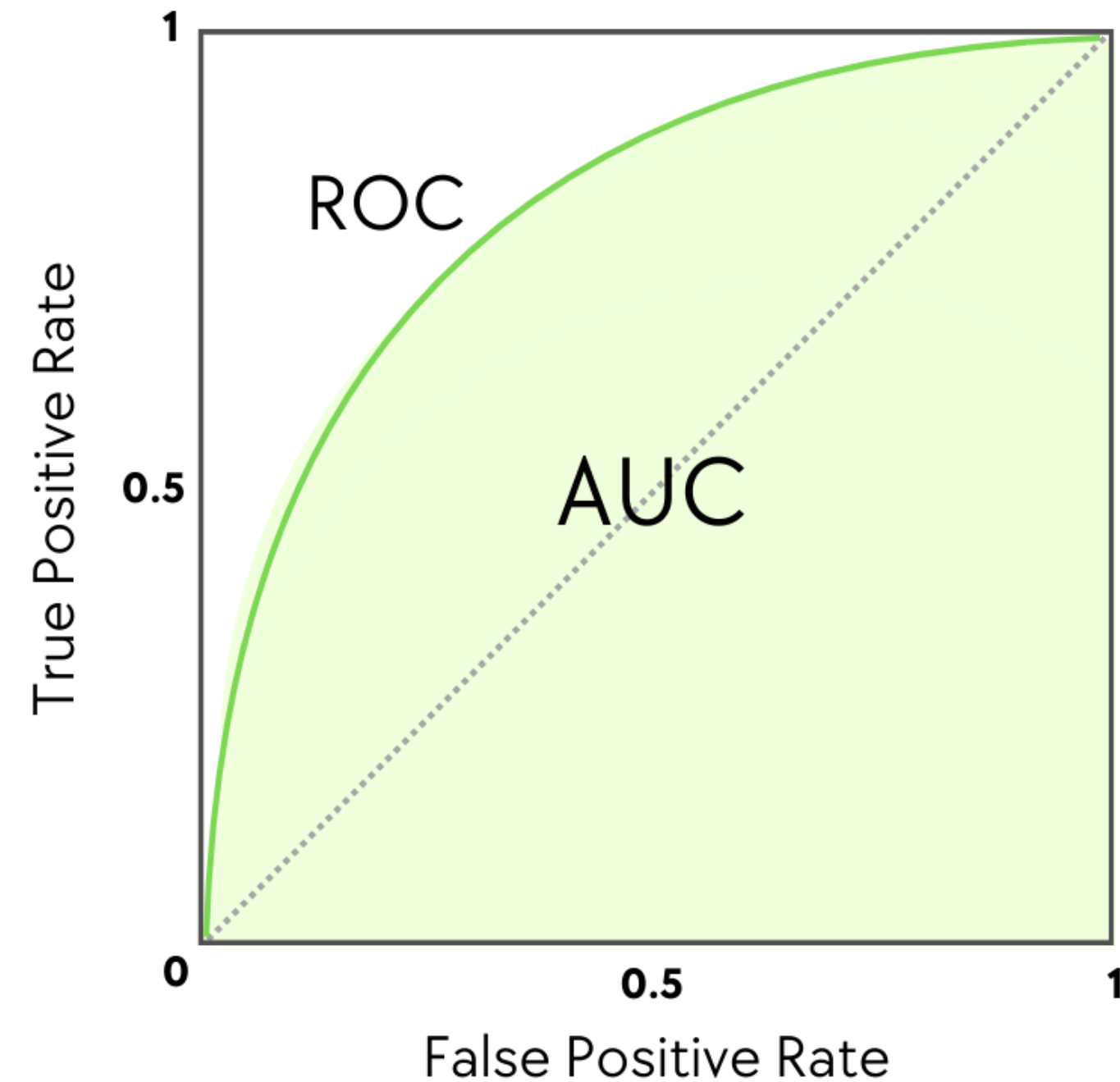
NEW EVALUATION TECHNIQUE



AUC (Area under curve)

Area under the ROC Curve.

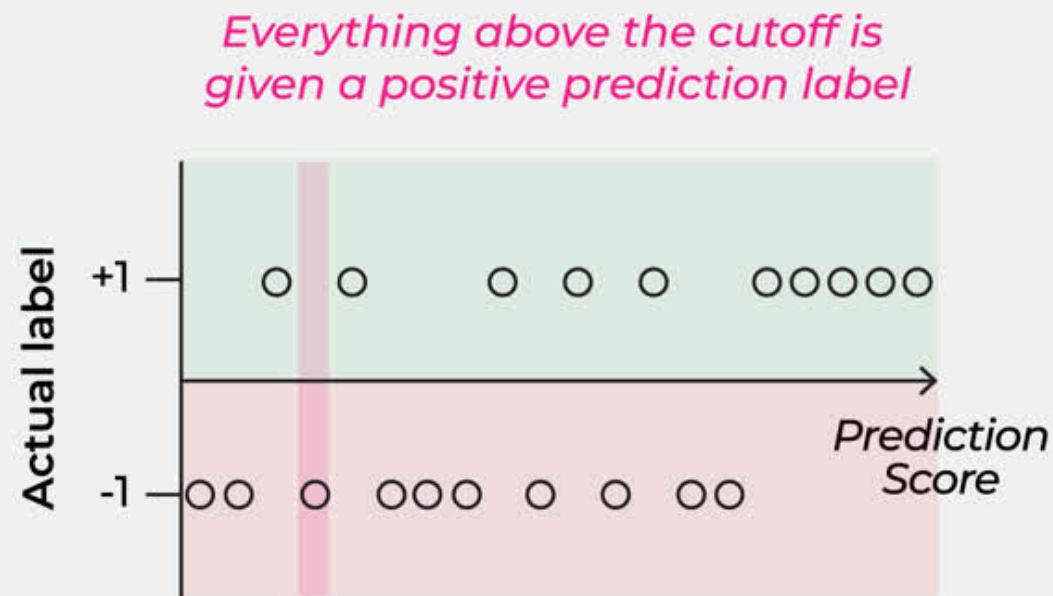
An AUC of 0.5 suggests random guessing, while 1.0 represents a perfect classifier.



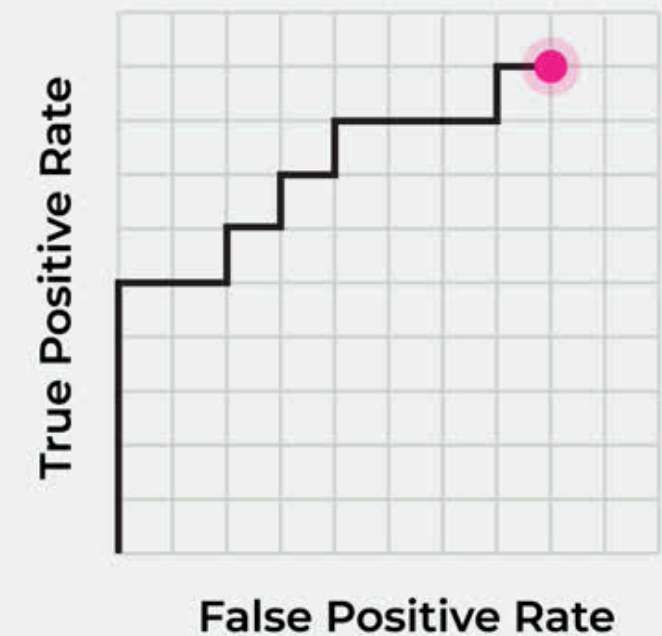
ROC AND AUC

True Positive Rate: 9/10

False Positive Rate: 8/10

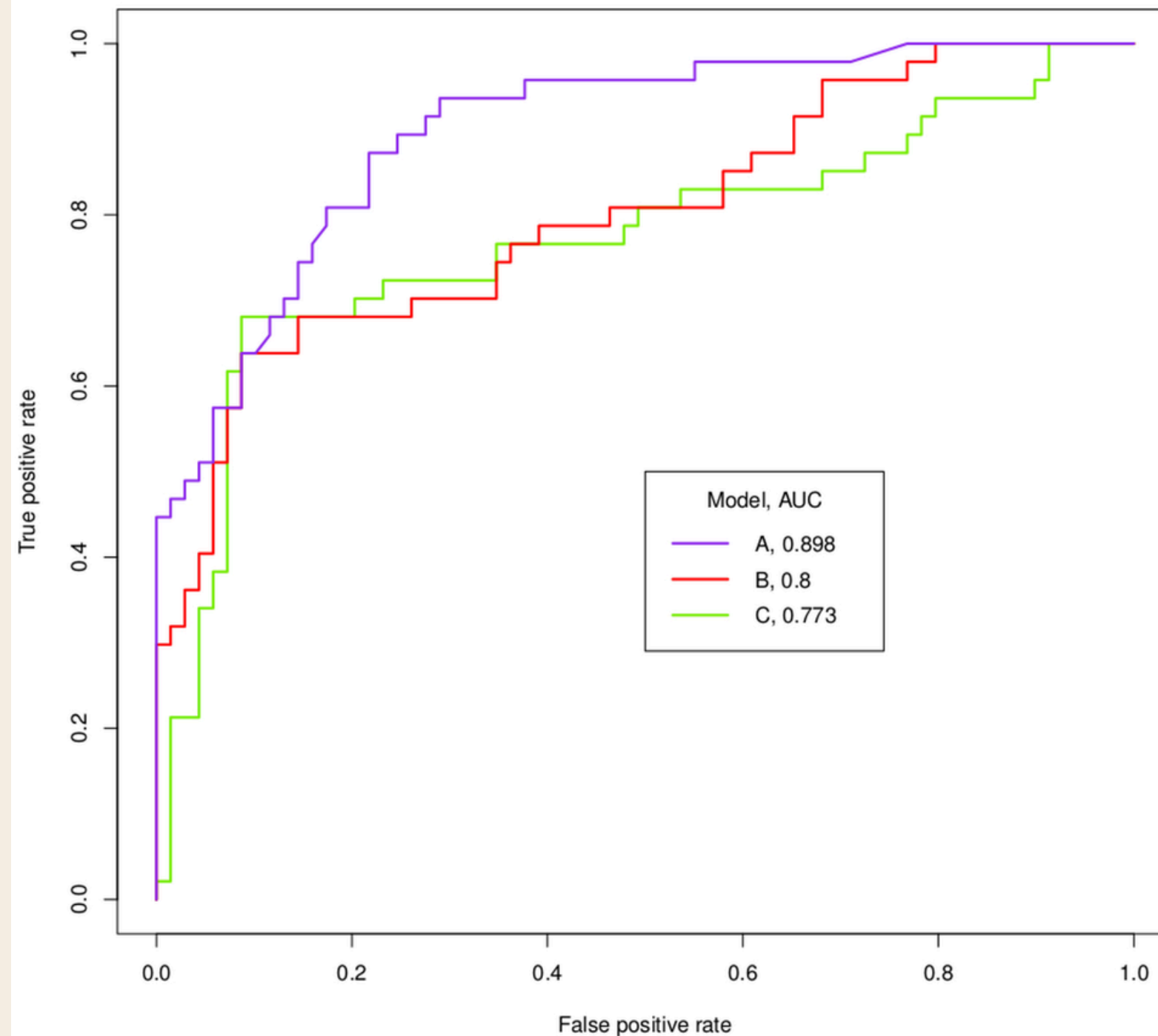


	Prediction label	
	POS (+)	NEG (-)
Actual label		
POS (+)	9	1
NEG (-)	8	2



View GIF here! <https://arize.com/blog/what-is-auc/>

ROC-AUC – VERY GOOD FOR COMPARING MODELS



which is very important for your
upcoming Assignment 2!

ROC-AUC IN R

```
library(ROCR)

# Step 1: Train your model
model <- naiveBayes(Y ~ ., data = traindata)

# Step 2: Get predicted probabilities for positive class
pred_prob <- predict(model, testdata, type = "raw")
score <- pred_prob[, 2] # probability of positive class (2nd column)

# Step 3: Create ROCR prediction object
pred_obj <- prediction(score, testdata$Y) # (predicted scores, true labels)

# Step 4: Compute TPR and FPR
roc <- performance(pred_obj, measure = "tpr", x.measure = "fpr")

# Step 5: Plot ROC curve
plot(roc, col = "red")

# Step 6: Compute AUC
auc <- performance(pred_obj, measure = "auc")
auc@y.values[[1]] # extract AUC value
```

ON-SITE QUESTIONS

ON-SITE QUESTION

PART I: ON-SITE QUESTION

(Naive Bayes classifier)

Data set `Titanic.csv` provides information on the fate of passengers on the fatal maiden voyage of the ocean liner Titanic. It includes the variables: economic status (class), sex, age and survival. We will train a naive Bayes classifier using this data set, and predict survival.

1. Compute the probabilities $P(Y = 1)$ (survived) and $P(Y = 0)$ (did not survive).
2. Compute the conditional probabilities $P(X_i = x_i | Y = 1)$ and $P(X_i = x_i | Y = 0)$, where $i = 1, 2, 3$ for the feature variables $X = \{class, sex, age\}$.

Discuss with your group! Time limit: 3.5 mins

ON-SITE PREP AND INSPECT

```
readLines("Data/Titanic.csv", n=3)  
  
titanic = read.csv("Data/Titanic.csv")  
  
head(titanic)  
str(titanic)
```

ON-SITE PREP AND INSPECT

```
> str(titanic)
'data.frame':   2201 obs. of  4 variables:
 $ Class      : chr  "3rd" "3rd" "3rd" "3rd" ...
 $ Sex        : chr  "Male" "Male" "Male" "Male" ...
 $ Age        : chr  "Child" "Child" "Child" "Child" ...
 $ Survived   : chr  "No" "No" "No" "No" ...
```

These are all categorical but defined as chr. Do we need to change it to factor?

You can, but don't really have to for Naive Bayes as it will treat character as categorical.

ON-SITE PREP AND INSPECT

```
> str(titanic)
'data.frame':   2201 obs. of  4 variables:
 $ Class      : chr  "3rd" "3rd" "3rd" "3rd" ...
 $ Sex        : chr  "Male" "Male" "Male" "Male" ...
 $ Age        : chr  "Child" "Child" "Child" "Child" ...
 $ Survived   : chr  "No" "No" "No" "No" ...
```

These are all categorical but defined as chr. Do we need to change it to factor?

But if it's categorical and defined as numerical (eg. 0,1). Then yes you have to convert

ON-SITE QUESTION 1

Q: Compute the probabilities $P(Y = 1)$ (survived) and $P(Y = 0)$ (did not survive).

```
surv.prop = prop.table(table(titanic$Survived))  
surv.prop
```

```
> table(titanic$Survived)
```

No	Yes
1490	711

```
> surv.prop
```

No	Yes
0.676965	0.323035

$P(Y=1) = 0.323$. $P(Y=0) = 0.677$

ON-SITE QUESTION 2

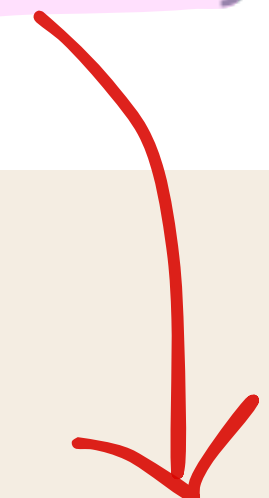
Q: Compute the conditional probabilities $P(X_i = x_i | Y = 1)$ and $P(X_i = x_i | Y = 0)$, where $i = 1, 2, 3$ for the feature variables $X = \{\text{class, sex, age}\}$

For $X = \text{class}$:

```
class.prop = prop.table(table(titanic[,c("Survived", "Class")]), margin = 1)  
class.prop
```

```
> table(titanic[,c("Survived", "Class")])
```

	Class			
Survived	1st	2nd	3rd	Crew
No	122	167	528	673
Yes	203	118	178	212



What does
margin = 1
mean?

ON-SITE QUESTION 2

Q: Compute the conditional probabilities $P(X_i = x_i | Y = 1)$ and $P(X_i = x_i | Y = 0)$, where $i = 1, 2, 3$ for the feature variables $X = \{\text{class, sex, age}\}$

For $X = \text{class}$:

```
class.prop = prop.table(table(titanic[,c("Survived", "Class")] ), margin = 1)
class.prop
```

```
> class.prop
      Class
Survived  1st      2nd      3rd      Crew
No  0.08187919 0.11208054 0.35436242 0.45167785
Yes 0.28551336 0.16596343 0.25035162 0.29817159
```

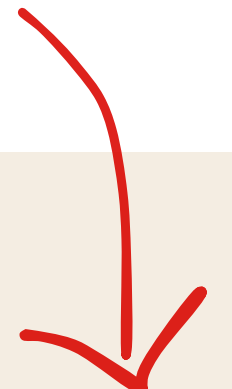


table of **row**
proportions

sum of each row = 1

ON-SITE QUESTION 2

Q: Compute the conditional probabilities $P(X_i = x_i | Y = 1)$ and $P(X_i = x_i | Y = 0)$, where $i = 1, 2, 3$ for the feature variables $X = \{\text{class}, \text{sex}, \text{age}\}$

$$P(\text{Adult} | \text{No}) = \frac{P(\text{Adult} \cap \text{No})}{P(\text{No})} = \frac{\text{count}(\text{Adult} \cap \text{No})/N}{\text{count}(\text{No})/N}$$

ON-SITE QUESTION 2

Q: Compute the conditional probabilities $P(X_i = x_i | Y = 1)$ and $P(X_i = x_i | Y = 0)$, where $i = 1, 2, 3$ for the feature variables $X = \{\text{class, sex, age}\}$

For $X = \text{class}$:

```
"  
P(Class = 1st | Survived = No) = 0.08187919  
P(Class = 2nd | Survived = No) = 0.11208054  
P(Class = 3rd | Survived = No) = 0.35436242  
P(Class = Crew | Survived = No) = 0.45167785  
  
P(Class = 1st | Survived = Yes) = 0.28551336  
P(Class = 2nd | Survived = Yes) = 0.16596343  
P(Class = 3rd | Survived = Yes) = 0.25035162  
P(Class = Crew | Survived = Yes) = 0.29817159  
"
```

ON-SITE QUESTION 2

Q: Compute the conditional probabilities $P(X_i = x_i | Y = 1)$ and $P(X_i = x_i | Y = 0)$, where $i = 1, 2, 3$ for the feature variables $X = \{\text{class}, \text{sex}, \text{age}\}$

For $X = \text{sex}$:

```
sex.prop = prop.table(table(titanic[,c("Survived", "Sex")] ), margin = 1)  
sex.prop
```

Sex		
Survived	Female	Male
No	0.08456376	0.91543624
Yes	0.48382560	0.51617440

ON-SITE QUESTION 2

Q: Compute the conditional probabilities $P(X_i = x_i | Y = 1)$ and $P(X_i = x_i | Y = 0)$, where $i = 1, 2, 3$ for the feature variables $X = \{\text{class}, \text{sex}, \text{age}\}$

For $X = \text{sex}$:

```
"  
P(Sex = Female | Survived = No)   = 0.08456376  
P(Sex = Male    | Survived = No)   = 0.91543624  
  
P(Sex = Female | Survived = Yes)  = 0.48382560  
P(Sex = Male   | Survived = Yes)  = 0.51617440  
"
```

ON-SITE QUESTION 2

Q: Compute the conditional probabilities $P(X_i = x_i | Y = 1)$ and $P(X_i = x_i | Y = 0)$, where $i = 1, 2, 3$ for the feature variables $X = \{\text{class}, \text{sex}, \text{age}\}$

For $X = \text{age}$:

```
age.prop = prop.table(table(titanic[,c("Survived", "Age")]), margin = 1)
age.prop
```

	Age	
Survived	Adult	Child
No	0.96510067	0.03489933
Yes	0.91983122	0.08016878

ON-SITE QUESTION 2

Q: Compute the conditional probabilities $P(X_i = x_i | Y = 1)$ and $P(X_i = x_i | Y = 0)$, where $i = 1, 2, 3$ for the feature variables $X = \{\text{class}, \text{sex}, \text{age}\}$

For $X = \text{age}$:

```
"  
P(Age = Adult | Survived = No)   = 0.96510067  
P(Age = Child | Survived = No)   = 0.03489933  
  
P(Age = Adult | Survived = Yes)  = 0.91983122  
P(Age = Child | Survived = Yes)  = 0.08016878  
"
```

ON-SITE QUESTION 3,4

(Naive Bayes classifier)

Data set `Titanic.csv` provides information on the fate of passengers on the fatal maiden voyage of the ocean liner Titanic. It includes the variables: economic status (class), sex, age and survival. We will train a naive Bayes classifier using this data set, and predict survival.

3. Predict survival for an adult female passenger in 2nd class cabin.
4. Compare your prediction in (c) with the one performed by the `naiveBayes()` function in package 'e1071'.

$$P(Y = y_j|X) \propto P(Y = y_j) \times \prod_{i=1}^m P(X_i = x_i|Y = y_j)$$

Discuss with your group! Time limit: 5 mins

ON-SITE QUESTION 3

Q: Predict survival for an adult female passenger in 2nd class cabin.

$$P(Y = y_j|X) \propto P(Y = y_j) \times \prod_{i=1}^m P(X_i = x_i|Y = y_j)$$

```
prob_surv <- surv.prop["Yes"] * (class.prop["Yes", "2nd"] *  
                                sex.prop["Yes", "Female"] *  
                                age.prop["Yes", "Adult"]) # 0.02385937  
  
prob_no_surv <- surv.prop["No"] * (class.prop["No", "2nd"] *  
                                   sex.prop["No", "Female"] *  
                                   age.prop["No", "Adult"]) # 0.006192319  
  
prob_surv / prob_no_surv # Ratio: 3.853059
```

ON-SITE QUESTION 4

Q: Compare your prediction in (c) with the one performed by the `naiveBayes()` function in package 'e1071'.

Step 0: Install and Load Package

```
# install.packages("e1071")  
library(e1071)
```

Step 1: Create Naive Bayes Model

```
M_NB <- naiveBayes(Survived ~ Class + Sex + Age, data = titanic)
```

ON-SITE QUESTION 4

Q: Compare your prediction in (c) with the one performed by the `naiveBayes()` function in package 'e1071'.

Step 3: Define test data (data you want to predict)

```
test <- data.frame(Class = "2nd", Sex = "Female", Age = "Adult")
```

Step 4: Predict the class of test data using Naive Bayes model

```
pred_class <- predict(M_NB, test)  
pred_class # Yes
```

ON-SITE QUESTION 4

Q: Compare your prediction in (c) with the one performed by the `naiveBayes()` function in package 'e1071'.

You can also get the probability by adding “raw”

```
pred_prob <- predict(M_NB, test, "raw")  
pred_prob
```

```
> pred_prob  
           No           Yes  
[1,] 0.2060556 0.7939444
```

Wait... This looks a little different from the probability in Question 3

ON-SITE QUESTION 4

It is same ratio as what we manually calculated above
(0.02385937, 0.006192319)! It's different because the package
normalized the values.

```
> pred_prob
```

	No	Yes
[1,]	0.2060556	0.7939444

No	Yes
0.2060556	0.7939444

Normalized process (just for your reference):

$$0.02385937 / (0.02385937 + 0.006192319) = 0.7939444$$
$$0.006192319 / (0.02385937 + 0.006192319) = 0.2060556$$

OFF-SITE QUESTIONS

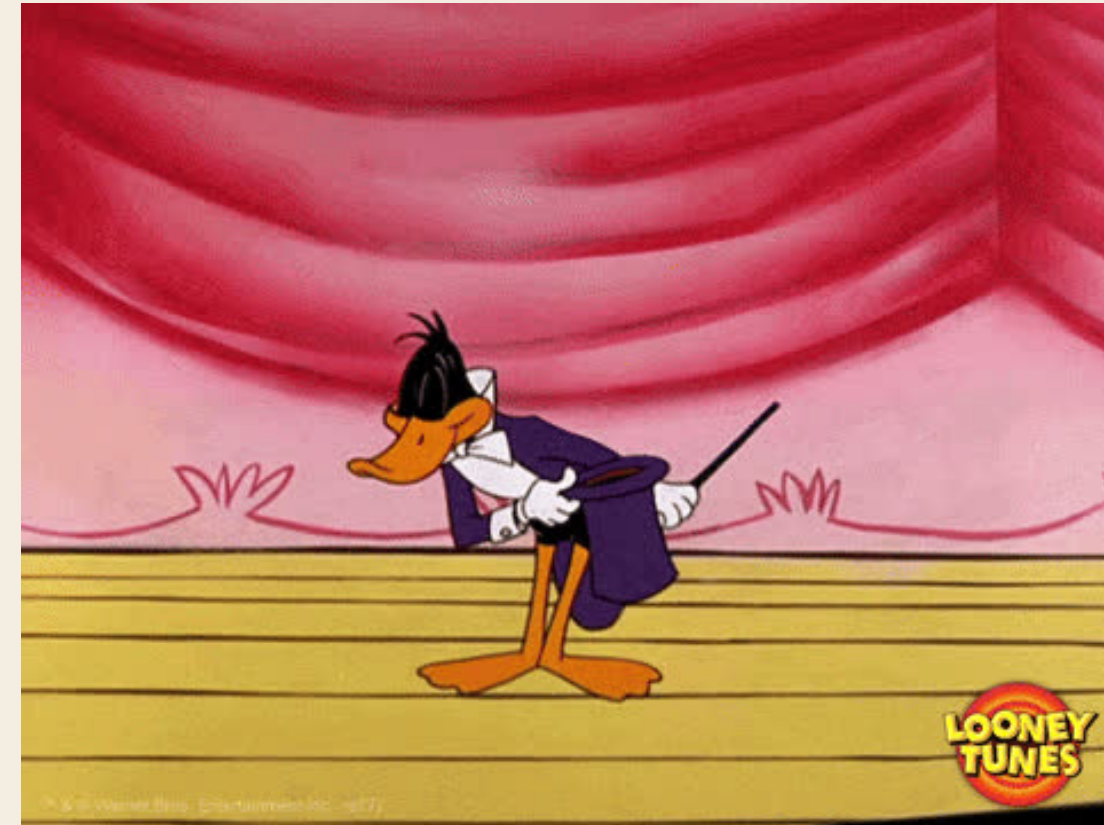
Let's go to code!

Extra Readings

Extra readings:

- **Types of Naive Bayes:** Multinomial Naive Bayes, Gaussian Naive Bayes
 - <http://analyticsvidhya.com/blog/2023/01/naive-bayes-algorithms-a-complete-guide-for-beginners/>
- **Videos of Naive Bayes (by StatQuest):**
 - Naive Bayes, Clearly Explained: <https://www.youtube.com/watch?v=O2L2Uv9pdDA>
 - Gaussian Naive Bayes, Clearly Explained: <https://www.youtube.com/watch?v=H3EjCKtlVog>

THANK YOU!



Weekly Slides & Code will be shared on Github (★ star it for bookmark):

<https://github.com/kr7001/DSA1101-teaching-materials>